SMARTFIT Inc.

TEST REPORT FOR

SMARTfit™ CPU Controller
Model: 3-56322

Tested To The Following Standard:

EN 300 220-2 v3.1.1

Report No.: 99819-9

Date of issue: June 2, 2017

ACCREDITED
Testing Certificates:
803.01, 803.02, 803.05, 803.06

This test report bears the accreditation symbol indicating that the testing performed herein meets the test and reporting requirements of ISO/IEC 17025 under the applicable scope of EMC testing for CKC Laboratories, Inc.

We strive to create long-term, trust based relationships by providing sound, adaptive, customer first testing services. We embrace each of our customers' unique EMC challenges, not as an interruption to set processes, but rather as the reason we are in business.

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Testing the Future
CKC LABORATORIES, INC.

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Test Report Information

REPORT PREPARED FOR:

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Camarillo, CA 93012

Representative: Jim Manley
Customer Reference Number: 2667

REPORT PREPARED BY:

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Mariposa, CA 95338

Project Number: 99819

DATE OF EQUIPMENT RECEIPT: May 17, 2017
DATE(S) OF TESTING: May 17-25, 2017

Report Authorization

The test data contained in this report documents the observed testing parameters pertaining to and are relevant for only the sample equipment tested in the agreed upon operational mode(s) and configuration(s) as identified herein. Compliance assessment remains the client's responsibility. This report may not be used to claim product endorsement by A2LA or any government agencies. This test report has been authorized for release under quality control from CKC Laboratories, Inc.

Steve Behm
Director of Quality Assurance & Engineering Services
CKC Laboratories, Inc.

Test Facility Information



Our laboratories are configured to effectively test a wide variety of product types. CKC utilizes first class test equipment, anechoic chambers, data acquisition and information services to create accurate, repeatable and affordable test results.

TEST LOCATION(S):
CKC Laboratories, Inc.
110 Olinda Place
Brea, CA 92823

Software Versions

CKC Laboratories Proprietary Software	Version
EMITest Emissions	5.03.02
EMITest Immunity	5.03.02

SUMMARY OF RESULTS

Standard / Specification: EN 300 220-2 v3.1.1

Test Procedure	Description	Modification	Results
4.2 All Equipment Conformance Requirements			
4.2.1	Operating Frequency	NA	Manufacturer declaration
4.2.2	Unwanted Emissions in the Spurious Domain	Mod. #1	Pass
4.3 Transmitter Conformance Requirements			
4.3.1	Effective Radiated Power	NA	Pass
4.3.2	Maximum E.R.P. Spectral Density	NA	NA8
4.3.3	Duty Cycle	NA	NA1
4.3.4	Occupied Bandwidth	NA	Pass
4.3.5	TX Out of Band Emissions	NA	Pass
4.3.6	Transient Power	NA	Pass
4.3.7	Adjacent Channel Power	NA	NA2
4.3.8	TX Behavior Under Low Voltage Conditions	NA	NA7
4.3.9	Adaptive Power Control	NA	NA3
4.3.10	FHSS Equipment	NA	NA4
4.3.11	Short Term Behaviour	NA	NA5
4.4 Receivers Conformance Requirements			
4.4.1	RX Sensitivity	NA	NA6
4.4.2	Blocking	NA	Pass
4.5 Polite Spectrum Access Conformance Requirements			
4.5.2	Clear Channel Assessment Method	NA	NA6
4.5.3	Polite Spectrum Access Timing Parameters	NA	NA6
4.5.4	Adaptive Frequency Agility	NA	NA6

NA = Not Applicable
NA1: Not applicable because there is no requirement. The EUT operates within band I.
NA2: Not applicable because the EUT's occupied bandwidth is more than 25kHz.
NA3: Not applicable because the EUT does not have adaptive power control.
NA4: Not applicable because the EUT is not a FHSS equipment.
NA5: Not applicable because the EUT does not operate in band where Ton or Toff limit are specified in annex C, table C.1
NA6: Not applicable because the EUT does not have polite spectrum access technique.
NA7: Not applicable because the EUT does not operate on battery power.
NA8: Not applicable because the EUT Modulation bandwidth is less than 250kHz.

Modifications During Testing

This list is a summary of the modifications made to the equipment during testing.

Summary of Conditions
Modification #1: Installed a ferrite (manufacturer: Fair-Rite, PN: 0431177081) on AC cable with two loops.

Modifications listed above must be incorporated into all production units.

Conditions During Testing

This list is a summary of the conditions noted to the equipment during testing.

Summary of Conditions
None

EQUIPMENT UNDER TEST

During testing numerous configurations may have been utilized. The configurations listed below support compliance to the standard(s) listed in the Summary of Results section.

Configuration 1

Equipment Tested:

Device	Manufacturer	Model #	S/N
SMARTfit™ CPU Controller	SMARTFIT Inc.	3-56322	NA

Support Equipment:

Device	Manufacturer	Model #	S/N
SMARTfit™ Pod Target	SMARTFIT Inc.	3-56228	NA
Power Supply	Triad Magnetics	WSU060-3000	NA

EN 300 220-2 V3.1.1

4.2 ALL EQUIPMENT CONFORMANCE REQUIREMENT

4.2.1 Operating Frequency

Test Setup/Conditions			
Test Location:	Brea	Test Engineer:	D. Nguyen
Manufacturer Declarations:			

Operational Frequency band or bands	433.05MHz - 434.79MHz
Nominal Operating Frequency or Frequencies	433.92MHz
Operating Channel width(s) - OCW	116kHz

4.2.2 Unwanted Emissions in the Spurious Domain

Test Setup/Conditions			
Test Location:	Brea Lab A	Test Engineer:	Don Nguyen
Test Method:	EN 300 220-1 v3.1.1 §5.9.3.3	Test Date(s):	5/17/2017
Declarations:	Modification #1 was in place during testing.		
Configuration:	1		
Test Setup:	The EUT is placed on Styrofoam platform at 1.5m in height. The input voltage is 230Vac/50Hz. USB and DC ports are unterminated and not accessible to user. The EUT is tested in preferred orientation declared by the manufacturer. The EUT is set to continuously transmit. Operating frequency: 433.92MHz Antenna gain: 0dBi Software output power: <1mW Test signal: D-M3 Frequency range of measurement = 9kHz-6GHz 9 kHz – 150 kHz, RBW=1kHz, VBW=3kHz 150 kHz – 30 MHz, RBW=10 kHz, VBW=27 kHz 30 MHz – 1000 MHz, RBW=100 kHz, VBW=300 kHz 1000 MHz – 6000 MHz, RBW=1 MHz, VBW=3 MHz		

Environmental Conditions			
Ambient Temperature (ºC)	18	Relative Humidity (%):	42

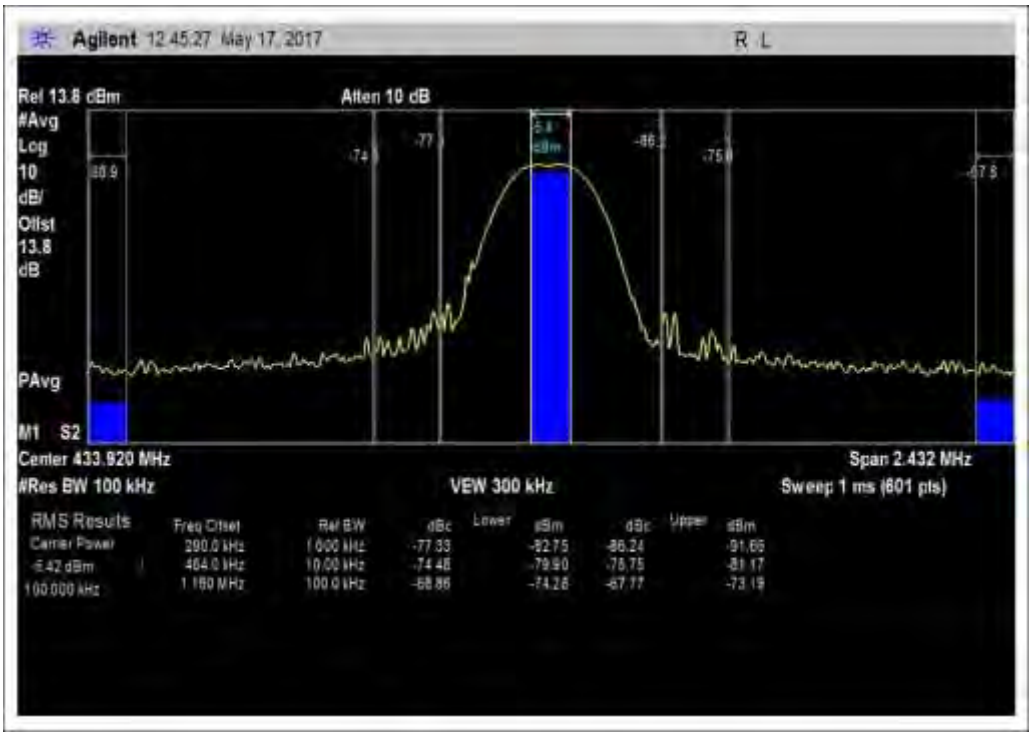
Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00309	Preamp	HP	8447D	3/14/2016	3/14/2018
01995	Biconilog Antenna	Chase	CBL6111C	5/10/2016	5/10/2018
P05275	Attenuator	Weinschel	1W	5/5/2016	5/5/2018
P05050	Cable	Pasternack	RG223/U	1/20/2017	1/20/2019
P05198	Cable	Belden	8268	12/7/2016	12/7/2018
02672	Spectrum Analyzer	Belden	E4446A	3/2/2017	3/2/2019
00786	Preamp	HP	83017A	5/9/2016	5/9/2018
00849	Horn Antenna	ETS	3115	3/4/2016	3/4/2018
02946	Cable	Astrolab Inc.	32022-2-2909K-36TC	11/2/2015	11/2/2017
P06661	Cable	Andrew	LDF1-50	5/6/2016	5/6/2018
03419	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017
P05497	Dipole Antenna	Schwarzbeck	UHAP	1/19/2017	1/19/2019
01646	Horn Antenna	ETS	3115	3/4/2016	3/4/2018
00314	Loop Antenna	EMCO	6502	5/20/2016	5/20/2018

Test Data

TX Mode				
Frequency (MHz)	**Antenna polarity**	**Measure power (dBm)**	**Limit (dBm)**	**Result**
58.73	Horiz	-70.4	-54	Pass
59.43	Horiz	-69.7	-54	Pass
60.03	Horiz	-69.4	-54	Pass
60.31	Vert	-59.1	-54	Pass
60.93	Horiz	-70.8	-54	Pass
61.01	Vert	-58.8	-54	Pass
68.51	Vert	-59.6	-54	Pass
73.41	Vert	-57.2	-54	Pass
74.01	Vert	-57.3	-36	Pass
74.73	Horiz	-73.1	-36	Pass
119.16	Horiz	-71.3	-36	Pass
867.758	Horiz	-42.5	-36	Pass
867.764	Vert	-43.2	-36	Pass
867.764	Vert	-41.2	-36	Pass
1301.677	Horiz	-41.5	-30	Pass
1301.707	Vert	-47.0	-30	Pass
1735.577	Vert	-43.4	-30	Pass
1735.58	Horiz	-39.1	-30	Pass
2169.447	Vert	-51.6	-30	Pass
2169.45	Horiz	-50.3	-30	Pass
2603.237	Horiz	-45.6	-30	Pass
2603.317	Vert	-41.7	-30	Pass
3037.223	Horiz	-57.2	-30	Pass
3037.237	Vert	-56.1	-30	Pass
3471.093	Horiz	-44.7	-30	Pass
3471.107	Vert	-43.2	-30	Pass
3904.943	Vert	-46.2	-30	Pass
3904.947	Horiz	-42.7	-30	Pass
4338.83	Vert	-50.7	-30	Pass
4338.867	Horiz	-51.5	-30	Pass
4772.717	Vert	-43.7	-30	Pass
4772.753	Horiz	-48.1	-30	Pass
5640.427	Horiz	-46.3	-30	Pass
5640.457	Vert	-50.0	-30	Pass

Standby Mode				
Frequency (MHz)	**Antenna polarity**	**Measure power (dBm)**	**Limit (dBm)**	**Result**
58.73	Horiz	-70.4	-57	Pass
59.43	Horiz	-69.7	-57	Pass
60.03	Horiz	-69.4	-57	Pass
60.31	Vert	-59.1	-57	Pass
60.93	Horiz	-70.8	-57	Pass
61.01	Vert	-58.8	-57	Pass
68.51	Vert	-59.6	-57	Pass
73.41	Vert	-57.2	-57	Pass
74.01	Vert	-57.3	-57	Pass
74.73	Horiz	-73.1	-57	Pass
119.16	Horiz	-71.3	-57	Pass

RX Mode				
Frequency (MHz)	**Antenna polarity**	**Measure power (dBm)**	**Limit (dBm)**	**Result**
58.73	Horiz	-70.4	-57	Pass
59.43	Horiz	-69.7	-57	Pass
60.03	Horiz	-69.4	-57	Pass
60.31	Vert	-59.1	-57	Pass
60.93	Horiz	-70.8	-57	Pass
61.01	Vert	-58.8	-57	Pass
68.51	Vert	-59.6	-57	Pass
73.41	Vert	-57.2	-57	Pass
74.01	Vert	-57.3	-57	Pass
74.73	Horiz	-73.1	-57	Pass
119.16	Horiz	-71.3	-57	Pass



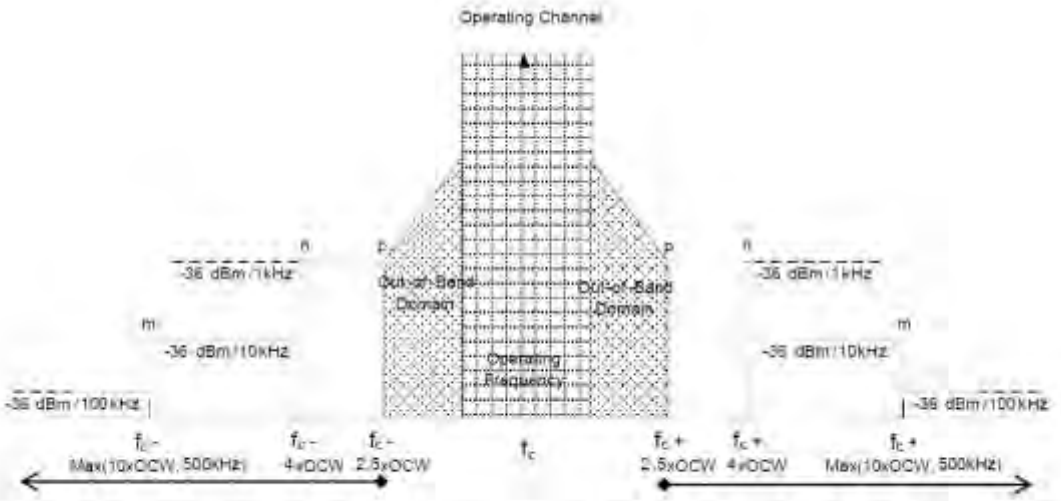


Figure 7: Spectrum Mask for Unwanted Emissions in the Spurious Domain with reference BW

State	47MHz - 74MHz 87.5MHz - 118MHz 174MHz - 230MHz 470MHz - 790MHz	Other frequencies below 1000MHz	Frequencies above 1000MHz
TX mode	-54 dBm	-36 dBm	-30 dBm
RX and all other modes	-57 dBm	-57 dBm	-47 dBm

Test Setup Photos









4.3 TRANSMITTER CONFORMANCE REQUIREMENT

4.3.1 Effective Radiated Power

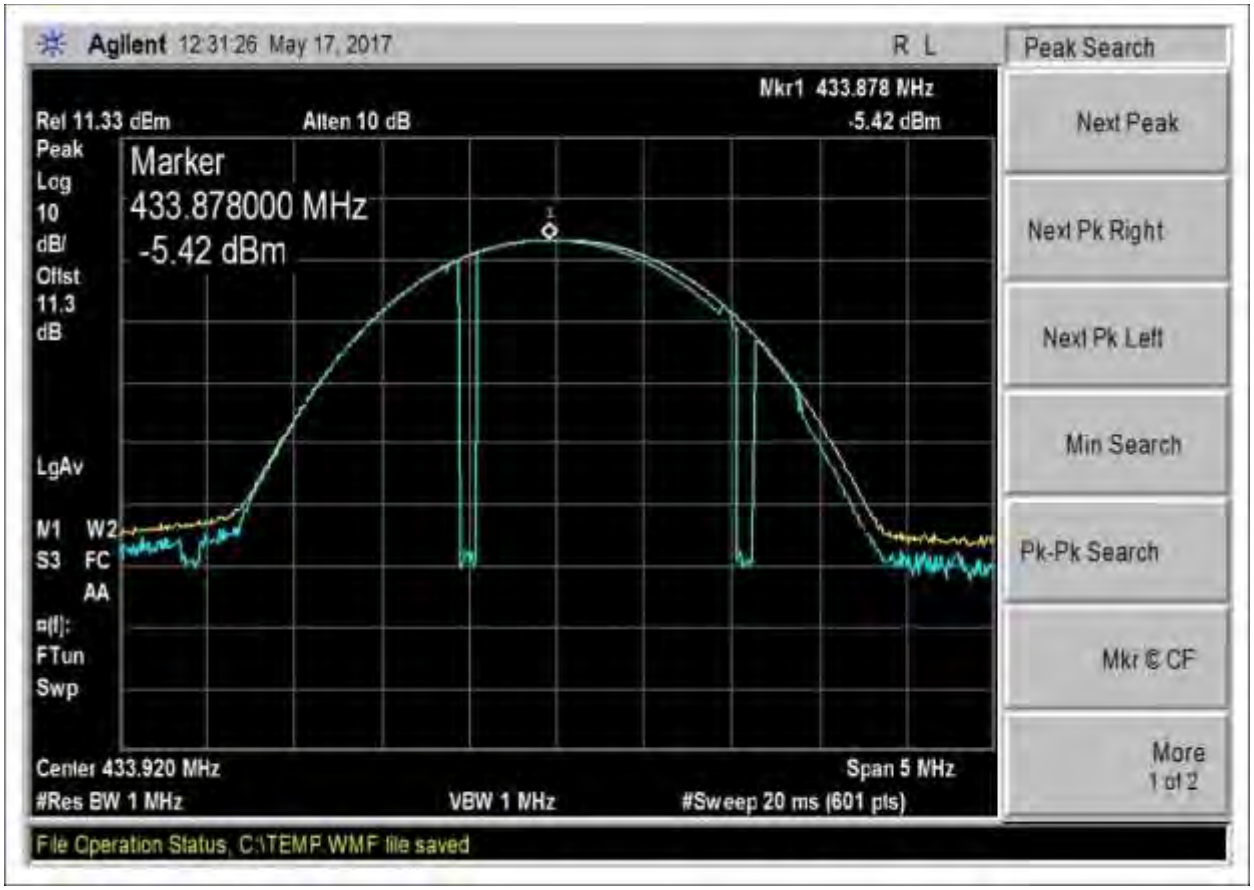
Test Setup/Conditions			
Test Location:	Brea Lab A	Test Engineer:	Don Nguyen
Test Method:	EN 300 220-1 v3.1.1 §5.2.2.1.2	Test Date(s):	5/17/2017
Declarations:	None		
Configuration:	1		
Test Setup:	The EUT is placed on Styrofoam platform at 1.5m in height. The input voltage is 230Vac/50Hz. USB and DC ports are unterminated and not accessible to user. The EUT is tested in preferred orientation declared by the manufacturer. The EUT is set to continuously transmit. Extreme condition test setup: The EUT is placed inside temperature chamber. The EUT top cover is removed to fit inside the temperature chamber. External AC power supply is connected to the EUT to vary input power. Probe antenna is placed next to the EUT to monitor fundamental transmission signal. Operating frequency: 433.92MHz Antenna gain: 0dBi Software output power: <1mW Test signal: D-M3 Frequency range of measurement = fundamental frequency RBW = 1 MHz, VBW = 1 MHz		

Environmental Conditions			
Ambient Temperature (ºC)	18	Relative Humidity (%):	42

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00309	Preamp	HP	8447D	3/14/2016	3/14/2018
01995	Biconilog Antenna	Chase	CBL6111C	5/10/2016	5/10/2018
P05275	Attenuator	Weinschel	1W	5/5/2016	5/5/2018
P05050	Cable	Pasternack	RG223/U	1/20/2017	1/20/2019
P05198	Cable	Belden	8268	12/7/2016	12/7/2018
02672	Spectrum Analyzer	Belden	E4446A	3/2/2017	3/2/2019
00786	Preamp	HP	83017A	5/9/2016	5/9/2018
03419	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017
P05497	Dipole Antenna	Schwarzbeck	UHAP	1/19/2017	1/19/2019
01878	Temperature Chamber	Thermotron Corp.	S 1.2 Mini-Max	5/15/2017	5/15/2019

Test Data

Test Conditions		Transmitter Power (mW)	Limit (mW)	Result
T_{nom}(25)°C	V_{nom}(230)Vac	0.29	1	Pass
T_{min}(-20)°C	V_{min}(253)Vac	0.14	1	Pass
	V_{max}(207)Vac	0.51	1	Pass
T_{max}(55)°C	V_{min}(253)Vac	0.04	1	Pass
	V_{max}(207Vac	0.02	1	Pass
Maximum Deviation		0.27	NA	NA



Test Setup Photos







Inside Temperature Chamber



Inside Temperature Chamber



Temperature Chamber

4.3.4 Occupied Bandwidth

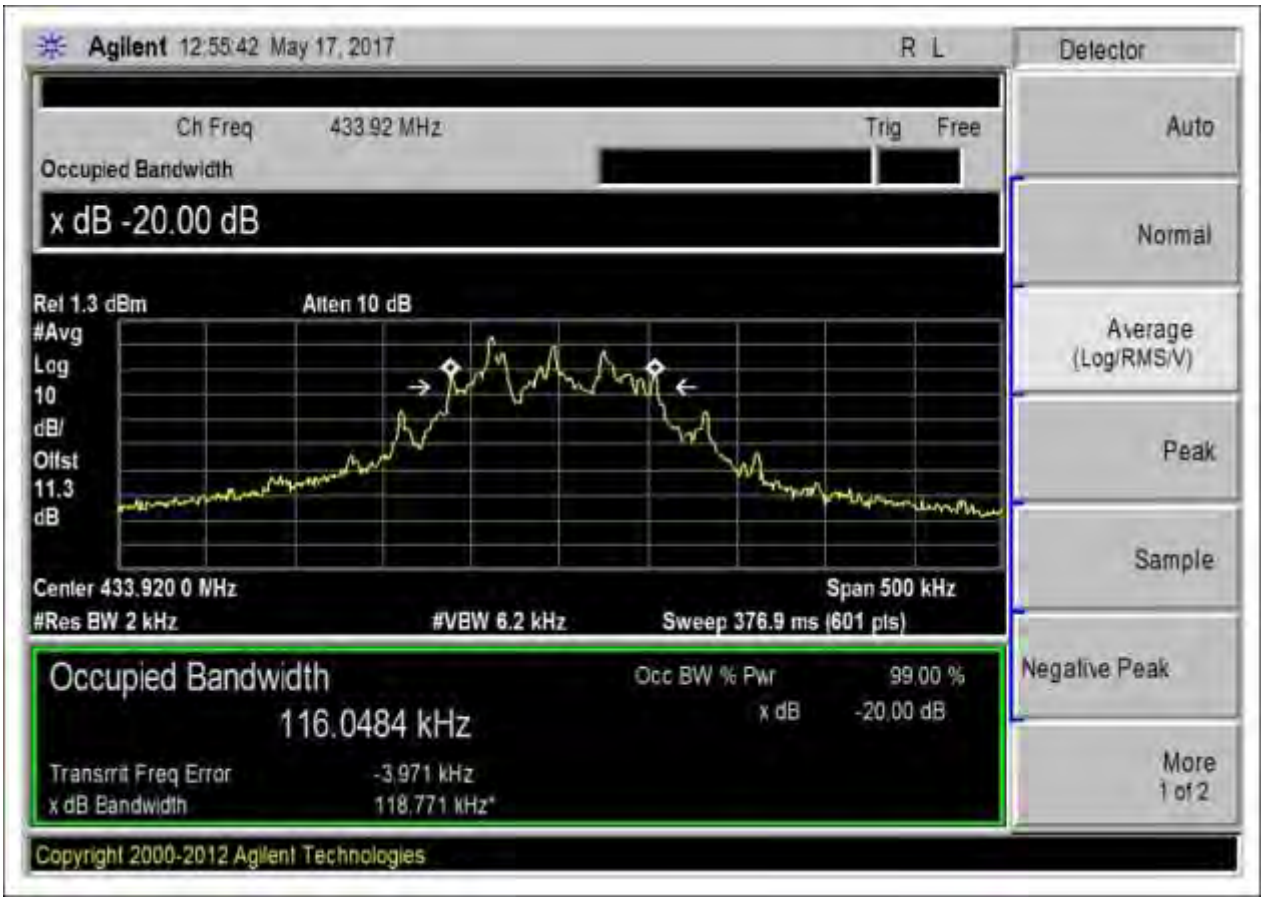
Test Setup/Conditions			
Test Location:	Brea Lab A	Test Engineer:	Don Nguyen
Test Method:	EN 300 220-1 v3.1.1 §5.6.3.4	Test Date(s):	5/17/2017
Declarations:	None		
Configuration:	1		
Test Setup:	The EUT is placed on Styrofoam platform at 1.5m in height. The input voltage is 230Vac/50Hz. USB and DC ports are unterminated and not accessible to user. The EUT is tested in preferred orientation declared by the manufacturer. The EUT is set to continuously transmit. Extreme condition test setup: The EUT is placed inside temperature chamber. The EUT top cover is removed to fit inside the temperature chamber. External AC power supply is connected to the EUT to vary input power. Probe antenna is placed next to the EUT to monitor fundamental transmission signal. Operating frequency: 433.92MHz Antenna gain: 0dBi Software output power: <1mW Test signal: D-M3 Frequency range of measurement = fundamental frequency RBW = 2 kHz, VBW = 6.2 kHz		

Environmental Conditions			
Ambient Temperature (ºC)	18	Relative Humidity (%):	42

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00309	Preamp	HP	8447D	3/14/2016	3/14/2018
01995	Biconilog Antenna	Chase	CBL6111C	5/10/2016	5/10/2018
P05275	Attenuator	Weinschel	1W	5/5/2016	5/5/2018
P05050	Cable	Pasternack	RG223/U	1/20/2017	1/20/2019
P05198	Cable	Belden	8268	12/7/2016	12/7/2018
02672	Spectrum Analyzer	Belden	E4446A	3/2/2017	3/2/2019
00786	Preamp	HP	83017A	5/9/2016	5/9/2018
01878	Temperature Chamber	Thermotron Corp.	S 1.2 Mini-Max	5/15/2017	5/15/2019

Test Data

Test Conditions		Occupied Bandwidth (kHz)	Limit	Result
T_{nom}(25)°C	V_{nom}(230)Vac	116.0484	Within Band	Pass
T_{min}(-20)°C	V_{min}(253)Vac	116.8767	Within Band	Pass
	V_{max}(207)Vac	117.7542	Within Band	Pass
T_{max}(55)°C	V_{min}(253)Vac	115.9157	Within Band	Pass
	V_{max}(207Vac	116.8380	Within Band	Pass
Maximum Deviation		1.7058	NA	NA



Test Setup Photos







Inside Temperature Chamber



Inside Temperature Chamber



Temperature Chamber

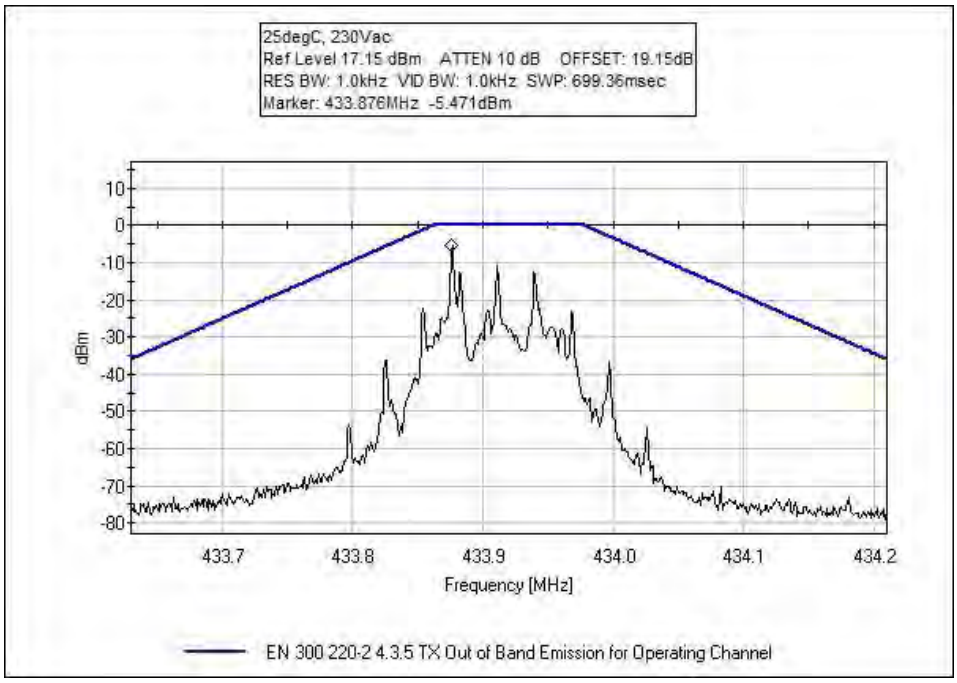
4.3.5 Transmitter Out of Band Emissions

Test Setup/Conditions			
Test Location:	Brea Lab A	Test Engineer:	Don Nguyen
Test Method:	EN 300 220-1 v3.1.1 §5.8.3.4	Test Date(s):	5/25/2017
Declarations:	None		
Configuration:	1		
Test Setup:	The EUT is placed on Styrofoam platform at 1.5m in height. The input voltage is 230Vac/50Hz. USB and DC ports are unterminated and not accessible to user. The EUT is tested in preferred orientation declared by the manufacturer. The EUT is set to continuously transmit. Extreme condition test setup: The EUT is placed inside temperature chamber. The EUT top cover is removed to fit inside the temperature chamber. External AC power supply is connected to the EUT to vary input power. Probe antenna is placed next to the EUT to monitor fundamental transmission signal. Operating frequency: 433.92MHz Antenna gain: 0dBi Software output power: <1mW Test signal: D-M3 Frequency range of measurement = fundamental frequency RBW = 1 kHz, VBW = 1 kHz **Result: pass**		

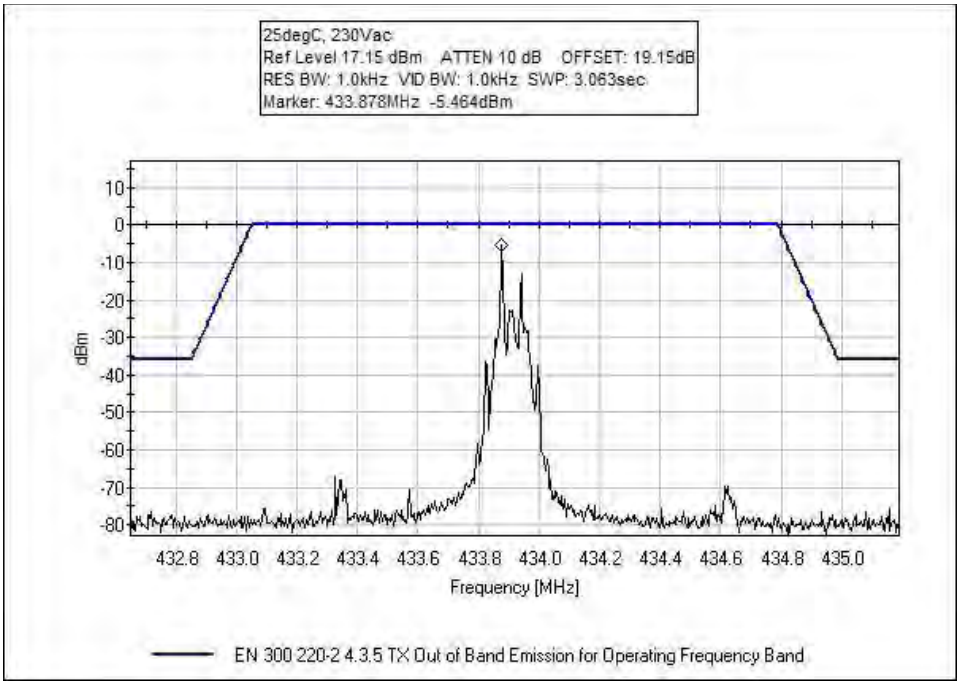
Environmental Conditions			
Ambient Temperature (ºC)	18	Relative Humidity (%):	42

Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00309	Preamp	HP	8447D	3/14/2016	3/14/2018
01995	Biconilog Antenna	Chase	CBL6111C	5/10/2016	5/10/2018
P05275	Attenuator	Weinschel	1W	5/5/2016	5/5/2018
P05050	Cable	Pasternack	RG223/U	1/20/2017	1/20/2019
P05198	Cable	Belden	8268	12/7/2016	12/7/2018
02672	Spectrum Analyzer	Belden	E4446A	3/2/2017	3/2/2019
00786	Preamp	HP	83017A	5/9/2016	5/9/2018
03419	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017
P05497	Dipole Antenna	Schwarzbeck	UHAP	1/19/2017	1/19/2019
01878	Temperature Chamber	Thermotron Corp.	S 1.2 Mini-Max	5/15/2017	5/15/2019

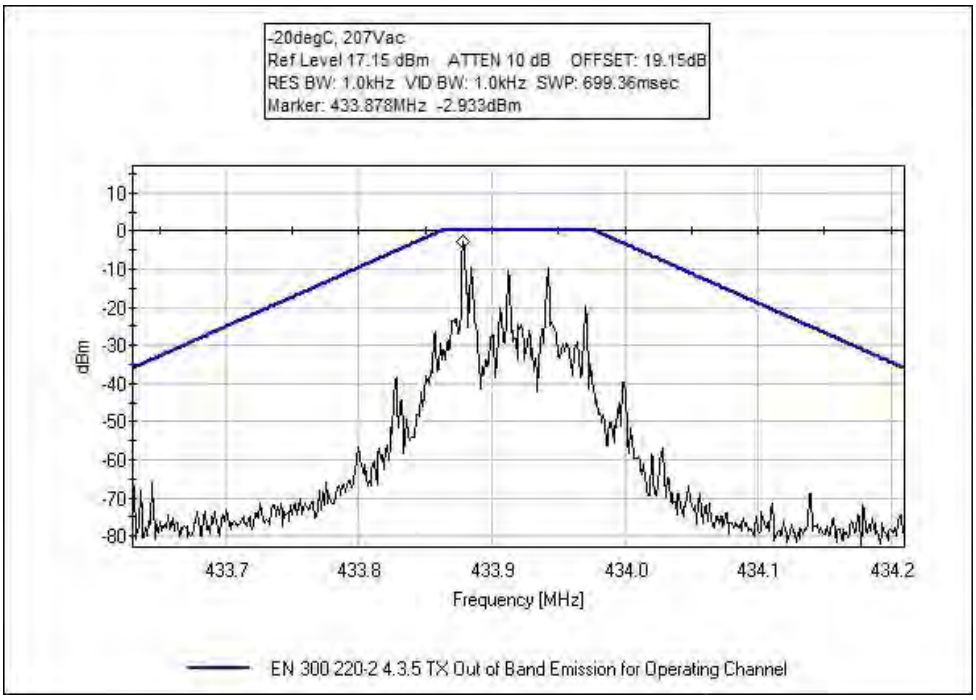
Test Data



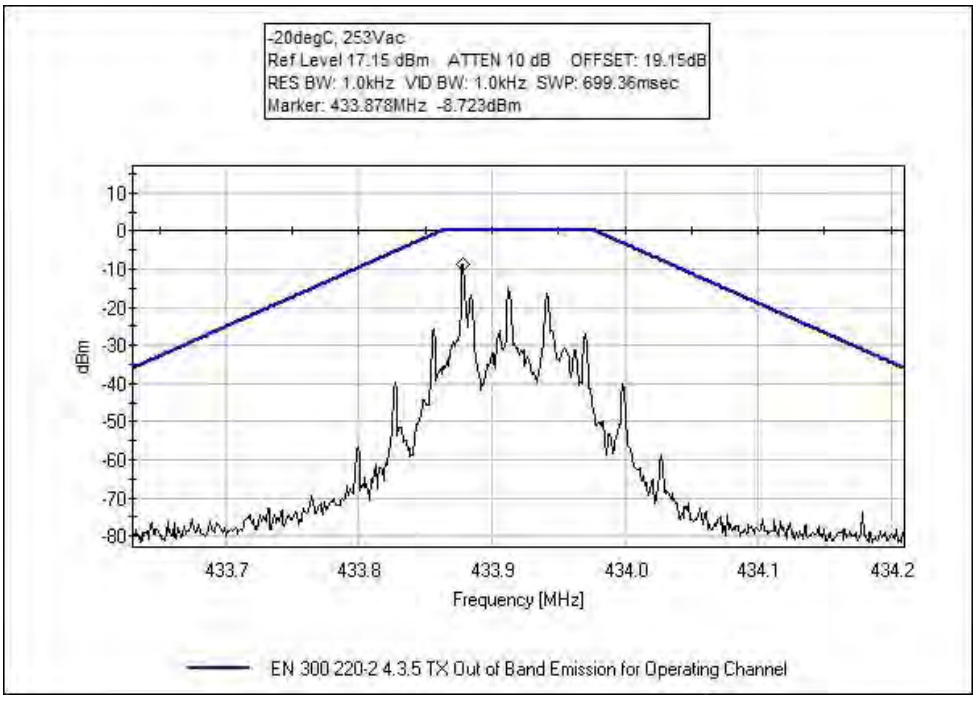
T_{nom}(25)°C, V_{nom}(230)Vac



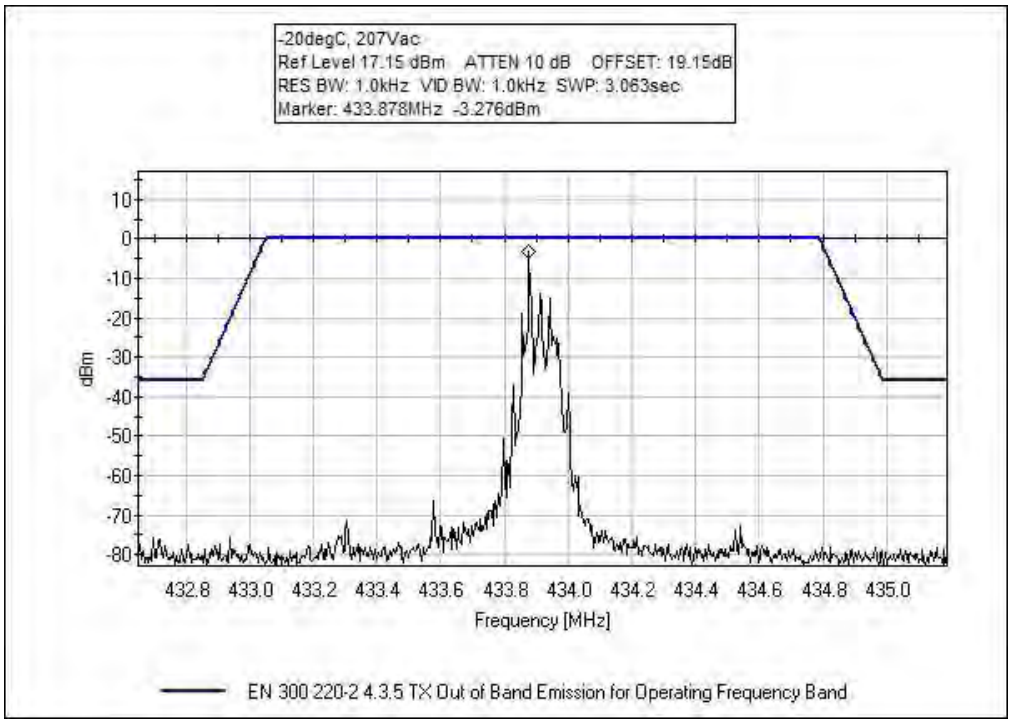
T_{nom}(25)°C, V_{nom}(230)Vac



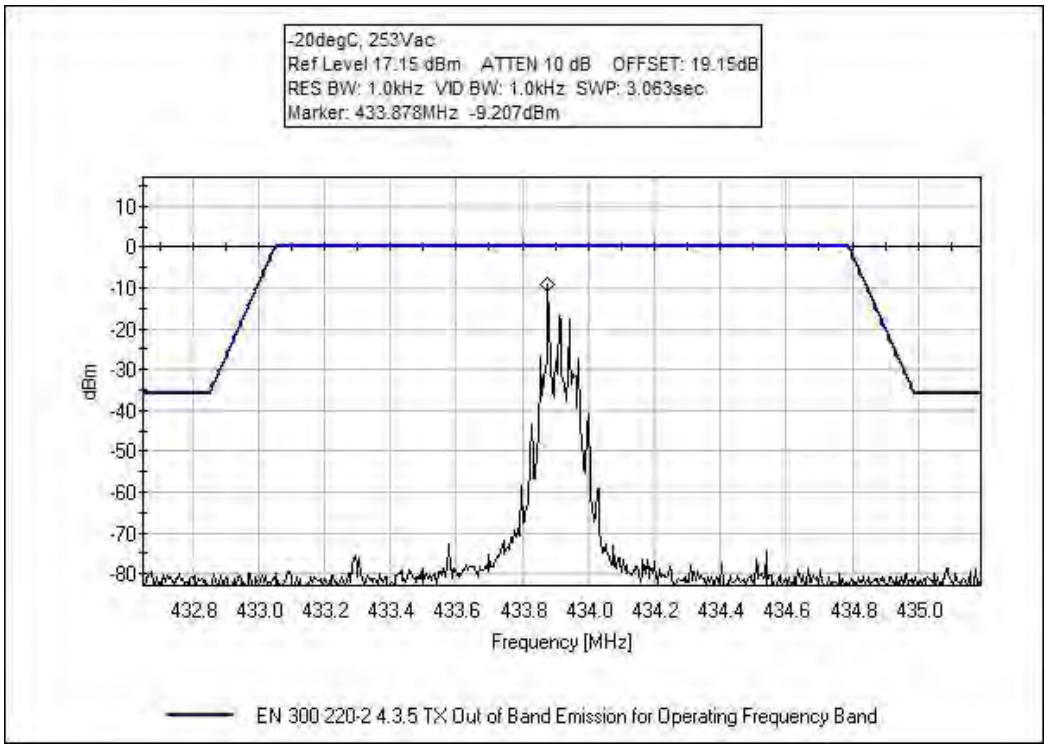
T_{low}(-20)°C, V_{low}(207)Vac



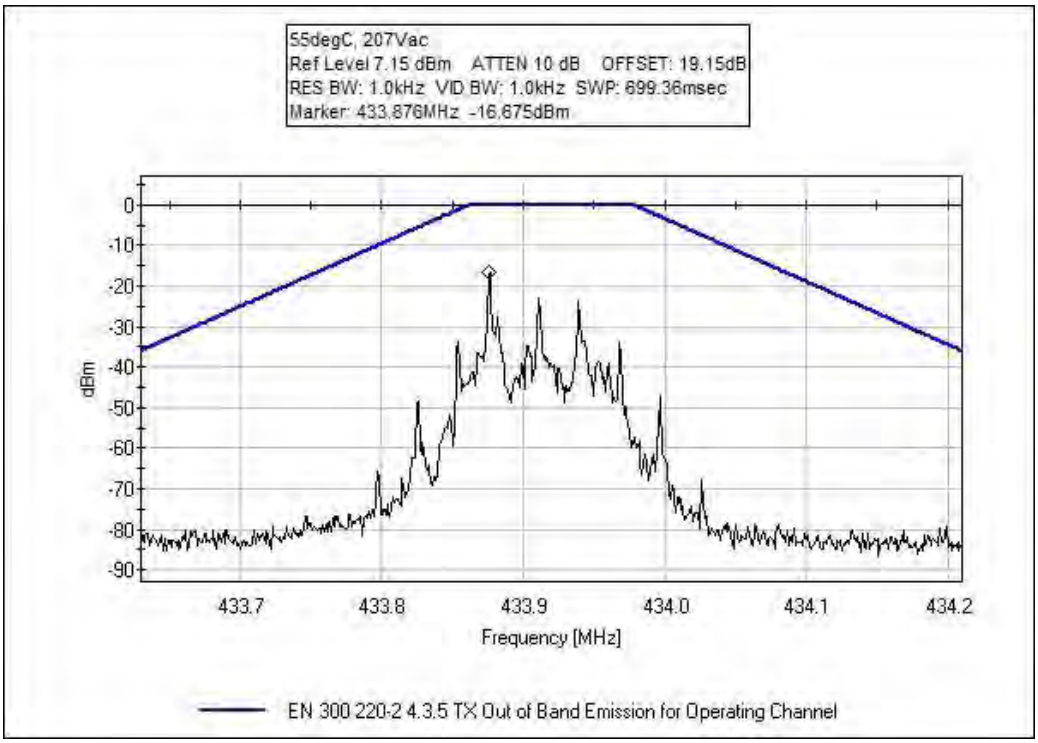
T_{low}(-20)°C, V_{hi}(253)Vac



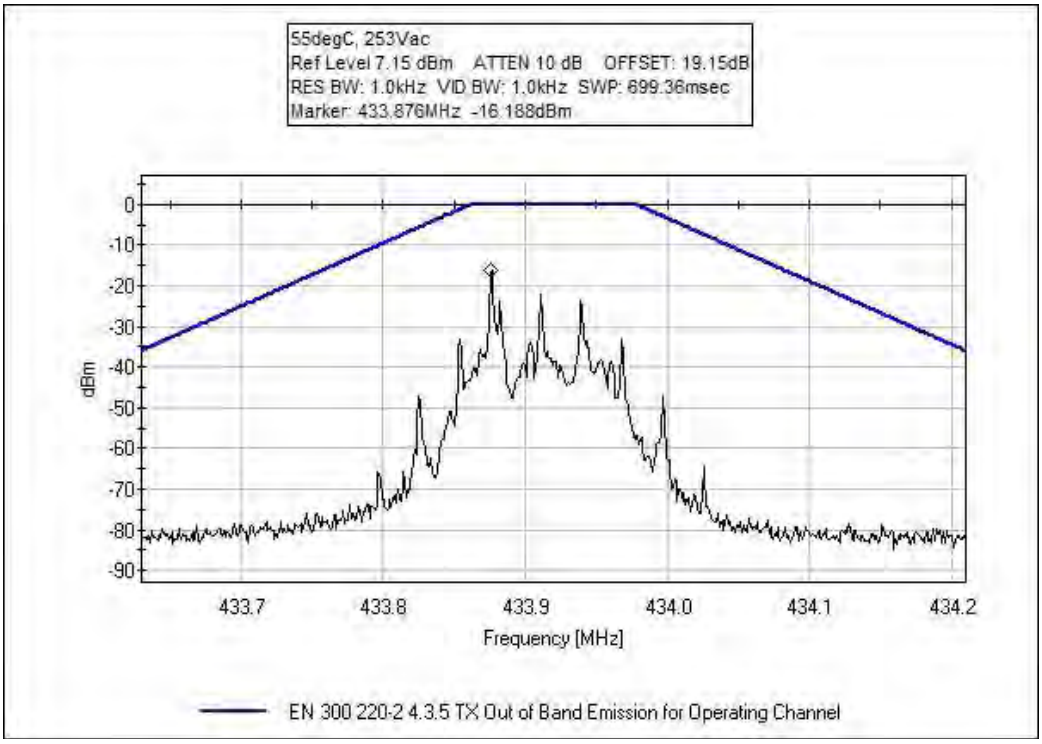
T_{low}(-20)°C, V_{low}(207)Vac



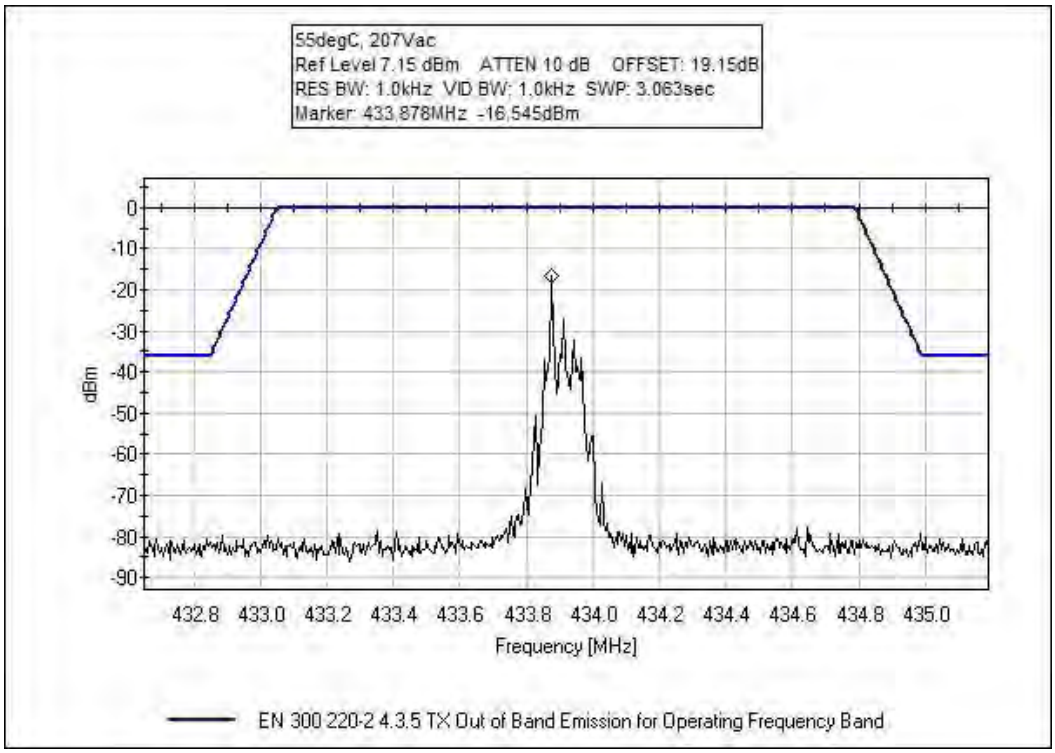
T_{low}(-20)°C, V_{hi}(253)Vac



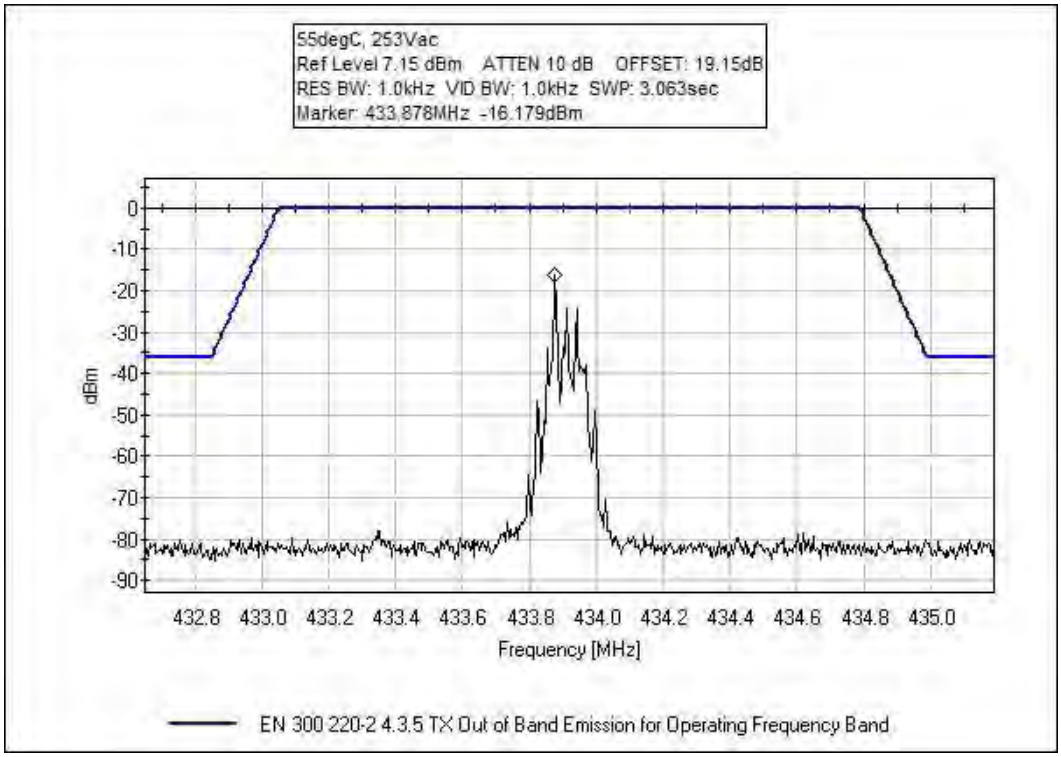
T_{low}(55)°C, V_{low}(207)Vac



T_{low}(55)°C, V_{hi}(253)Vac



T_{low}(55)°C, V_{low}(207)Vac



T_{low}(55)°C, V_{hi}(253)Vac

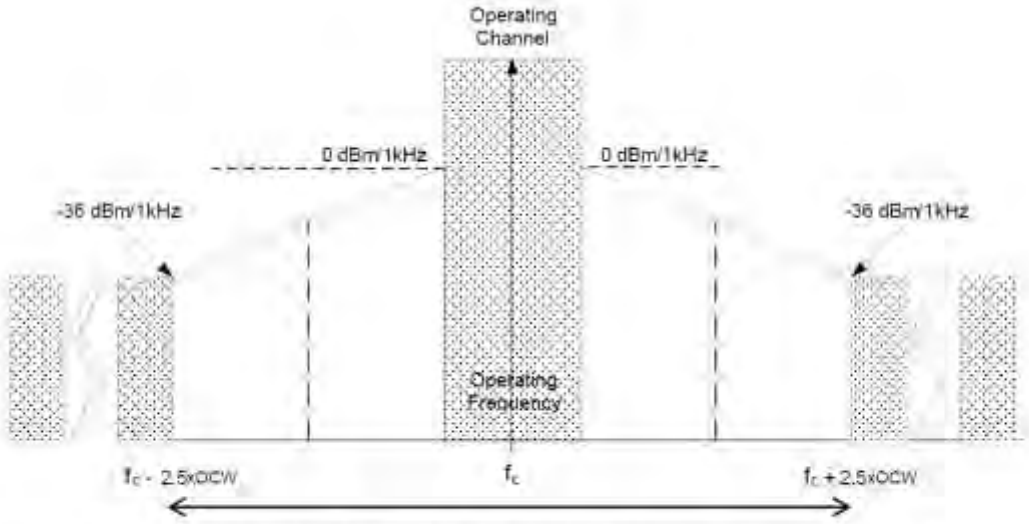


Figure 5: Out Of Band Domain for Operating Channel with reference BW

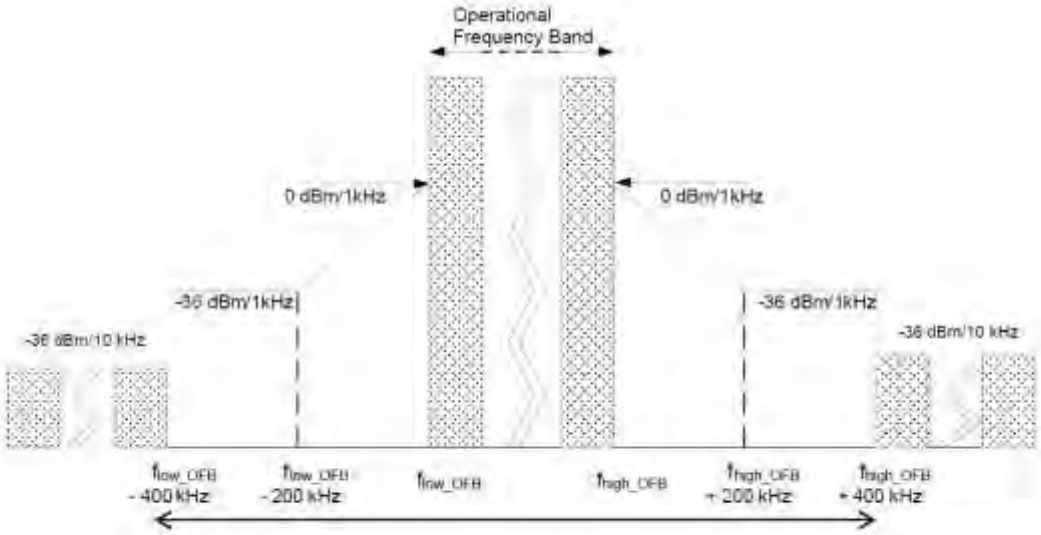


Figure 6: Out Of Band Domain for Operational Frequency Band with reference BW

Domain	Frequency Range	RBWREF	Max power limit
OOB limits applicable to Operational Frequency Band (See Figure 6)	f ≤ flow_OFB - 400 kHz	10 kHz	-36 dBm
	Flow_OFB - 400 kHz ≤ f ≤ flow_OFB - 200 kHz	1 kHz	-36 dBm
	flow - 200 kHz ≤ f < flow_OFB	1 kHz	See Figure 6
	f = flow_OFB	1 kHz	0 dBm
	f = fhigh_OFB	1 kHz	0 dBm
	Fhigh_OFB < f ≤ fhigh_OFB + 200 kHz	1 kHz	See Figure 6
	Fhigh_OFB + 200 kHz ≤ f ≤ fhigh_OFB + 400 kHz	1 kHz	-36 dBm
	Fhigh_OFB + 400 kHz ≤ f	10 kHz	-36 dBm
OOB limits applicable to Operating Channel (See Figure 5)	f = fc- 2.5 x OCW	1 kHz	-36 dBm
	fc - 2,5 x OCW ≤ f ≤ fc - 0,5 x OCW	1 kHz	See Figure 5
	f = fc - 0,5 x OCW	1 kHz	0 dBm
	f = fc + 0,5 x OCW	1 kHz	0 dBm
	fc + 0,5 x OCW ≤ f ≤ fc + 2,5 x OCW	1 kHz	See Figure 5
	f = fc+ 2,5 x OCW	1 kHz	-36 dBm
NOTE: f is the measurement frequency. fc is the Operating Frequency. Flow_OFB is the lower edge of the Operational Frequency Band. Fhigh_OFB is the upper edge of the Operational Frequency Band. OCW is the operating channel bandwidth.			

Test Setup Photos







Inside Temperature Chamber



Inside Temperature Chamber



Temperature Chamber

4.3.6 Transient Power

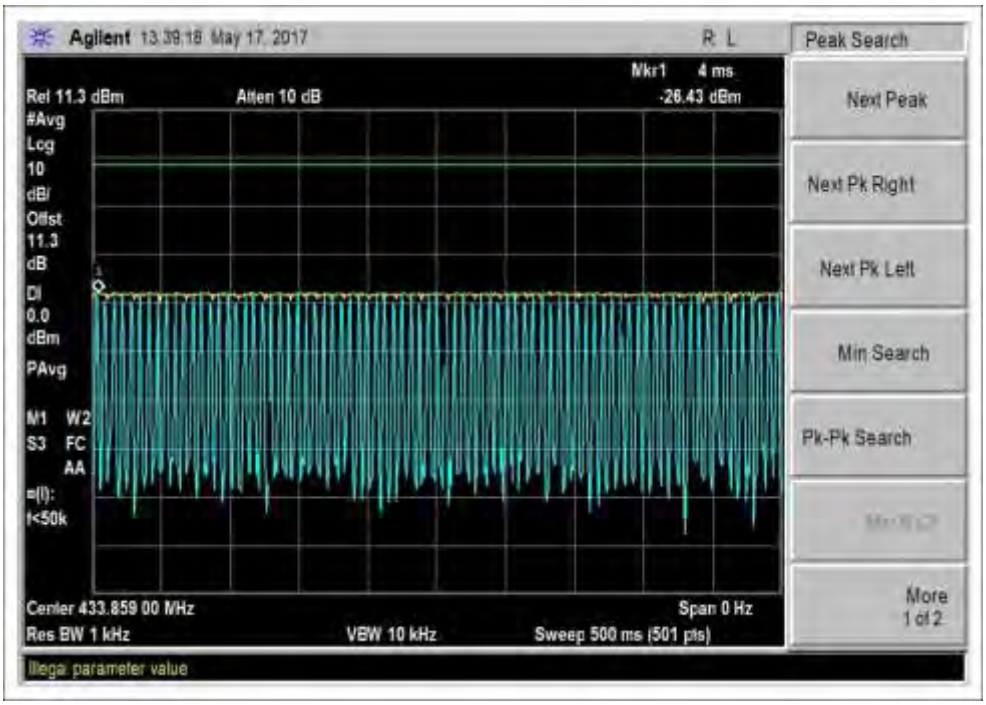
Test Setup/Conditions			
Test Location:	Brea Lab A	Test Engineer:	Don Nguyen
Test Method:	EN 300 220-1 v3.1.1 §5.10.3.2	Test Date(s):	5/17/2017
Declarations:	None		
Configuration:	1		
Test Setup:	The EUT is placed on Styrofoam platform at 1.5m in height. The input voltage is 230Vac/50Hz. USB and DC ports are unterminated and not accessible to user. The EUT is tested in preferred orientation declared by the manufacturer. The EUT is set to continuously transmit. Operating frequency: 433.92MHz Antenna gain: 0dBi Software output power: <1mW Test signal: D-M3 Frequency range of measurement = 433.92-0,5 x OCW -1200 kHz to 422.92+0,5 x OCW + 1200 kHz RBW = 1 kHz, VBW = 10 kHz (offset: +/-0,5 x OCW +/-3 kHz RBW = 10 kHz, VBW = 100 kHz (offset: +/-OCW) RBW = 100 kHz, VBW = 1000 kHz (offset: +/-0,5 x OCW +/- 400 kHz) RBW = 300 kHz, VBW = 3000 kHz (offset: +/-0,5 x OCW +/- 1200 kHz)		

Environmental Conditions			
Ambient Temperature (ºC)	18	Relative Humidity (%):	42

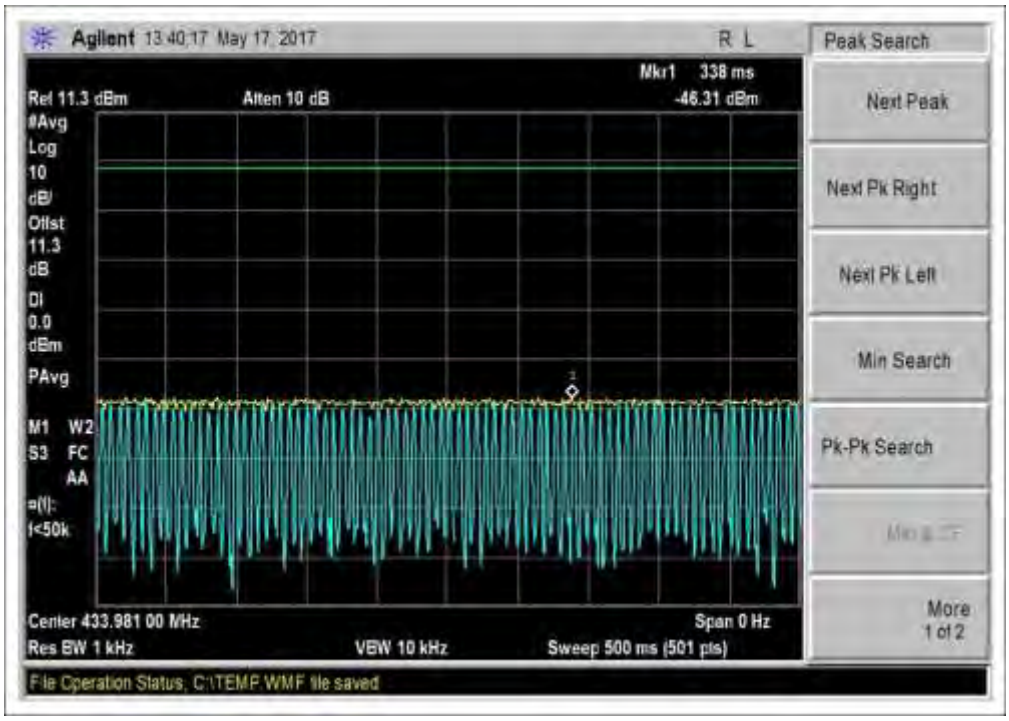
Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00309	Preamp	HP	8447D	3/14/2016	3/14/2018
01995	Biconilog Antenna	Chase	CBL6111C	5/10/2016	5/10/2018
P05275	Attenuator	Weinschel	1W	5/5/2016	5/5/2018
P05050	Cable	Pasternack	RG223/U	1/20/2017	1/20/2019
P05198	Cable	Belden	8268	12/7/2016	12/7/2018
02672	Spectrum Analyzer	Belden	E4446A	3/2/2017	3/2/2019
00786	Preamp	HP	83017A	5/9/2016	5/9/2018
03419	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017
P05497	Dipole Antenna	Schwarzbeck	UHAP	1/19/2017	1/19/2019

Test Data

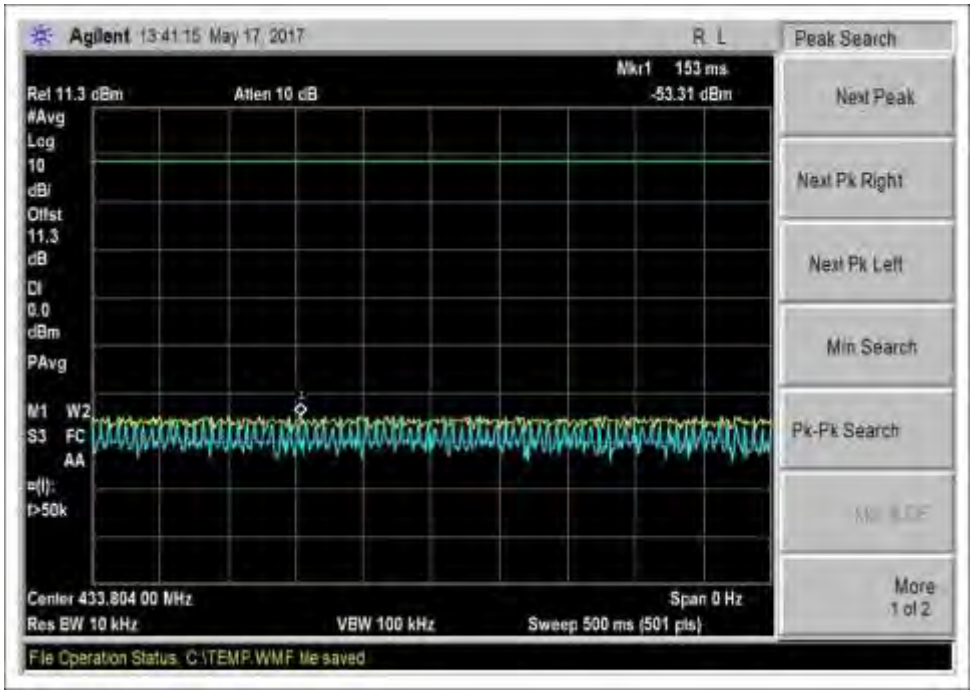
Offset Frequency (kHz)	Peak power (dBm)	Limit (dBm)	Result
-0,5 x OCW - 3 kHz	-26.43	0	Pass
+0,5 x OCW - 3 kHz	-46.31	0	Pass
-OCW	-53.31	0	Pass
+OCW	-54.82	0	Pass
-0,5 x OCW - 400 kHz	-68.83	-27	Pass
+0,5 x OCW + 400 kHz	-69.60	-27	Pass
-0,5 x OCW - 1200 kHz	-68.50	-27	Pass
+0,5 x OCW + 1200 kHz	-68.44	-27	Pass



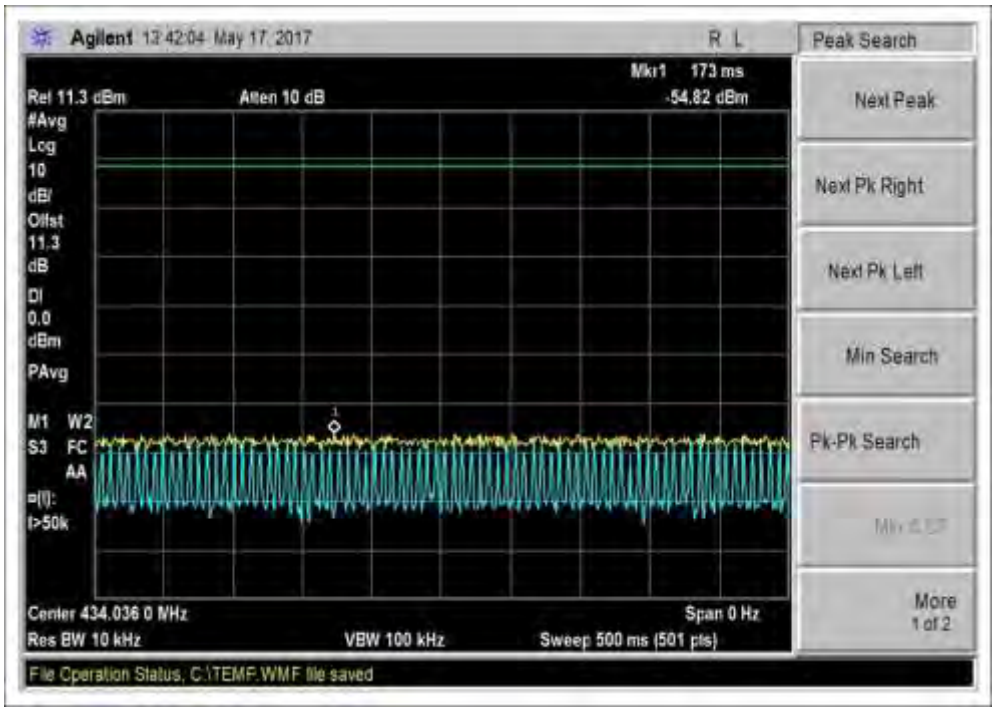
-0.5OCW-3kHz



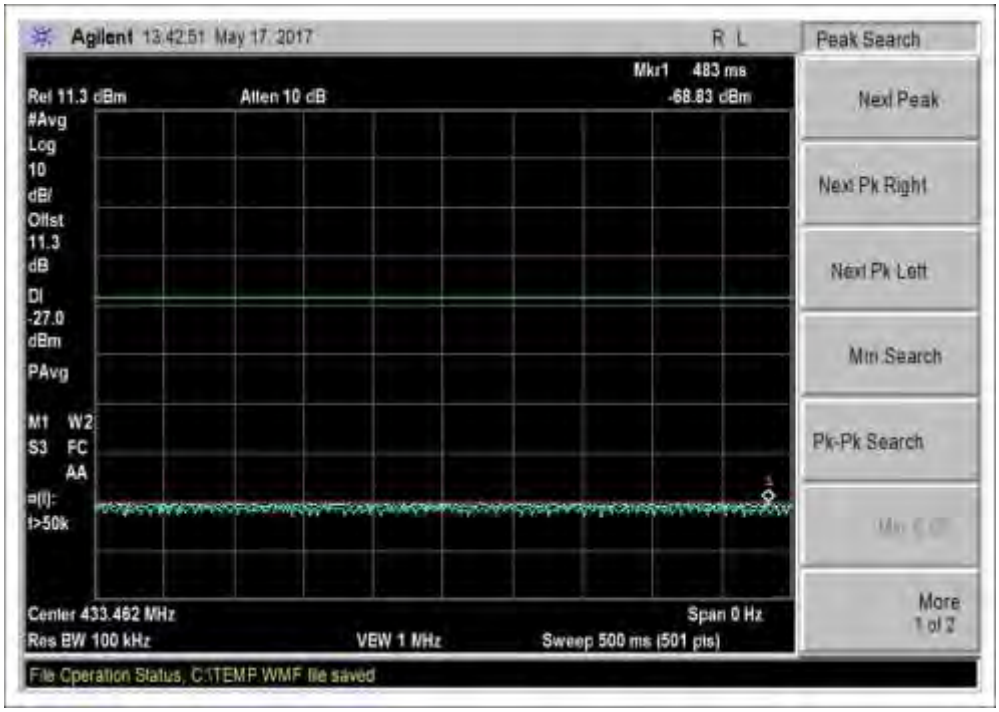
+0.5OCW+3kHz



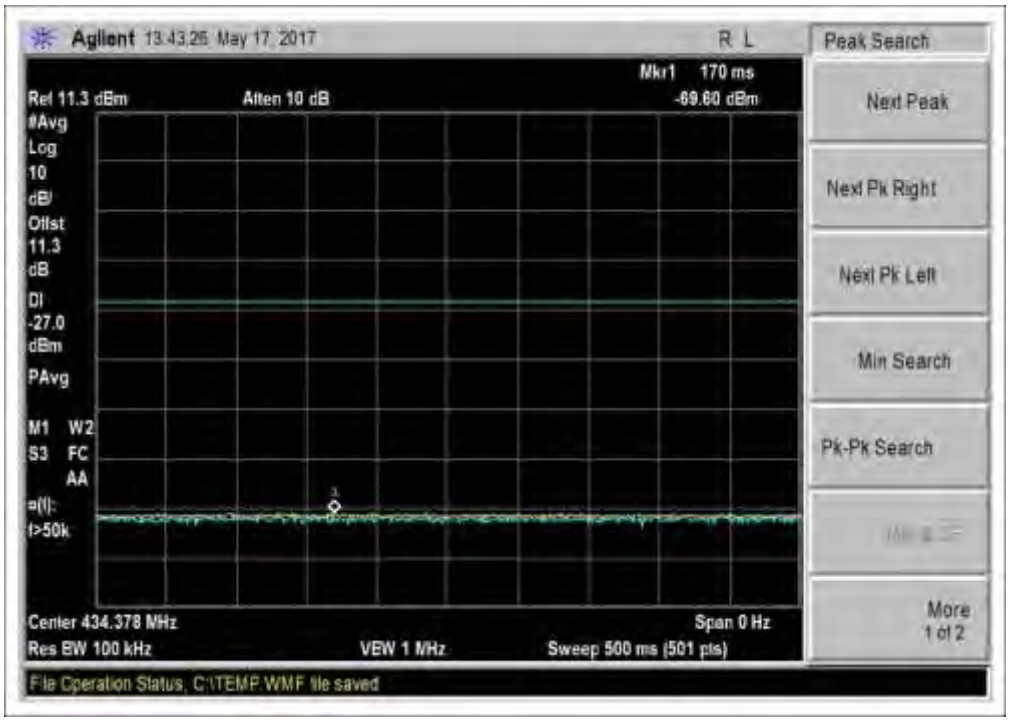
-OCW



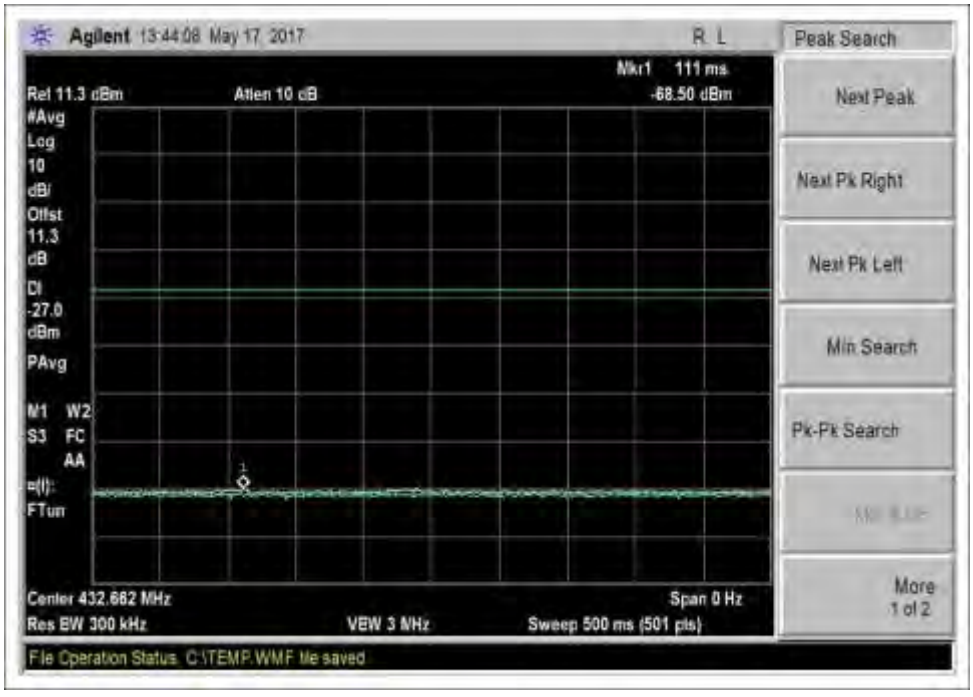
+OCW



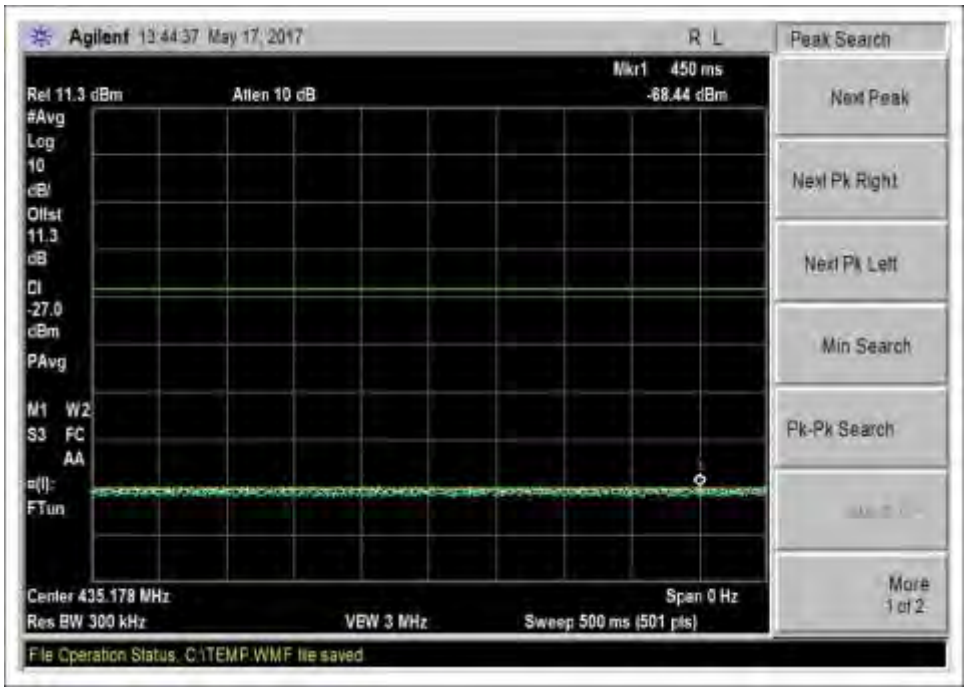
-0.5OCW-400kHz



+0.5OCW+400kHz



-0.5OCW-1200kHz



+0.5OCW+1200kHz

Test Setup Photos





4.4 RECEIVERS CONFORMANCE REQUIREMENT

4.4.2 Blocking

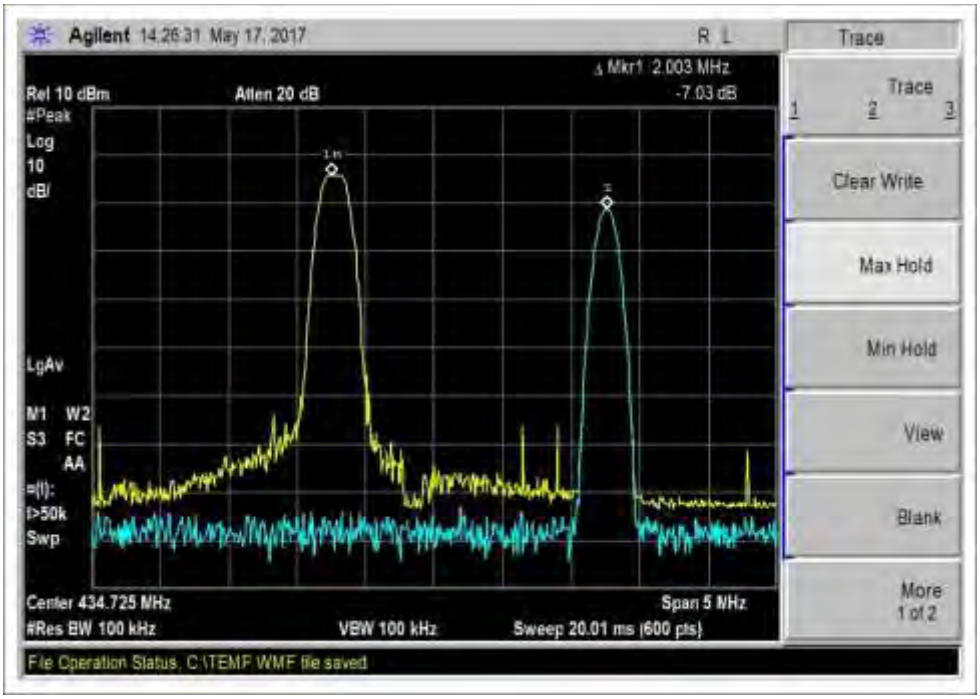
Test Setup/Conditions			
Test Location:	Brea Lab A	Test Engineer:	Don Nguyen
Test Method:	EN 300 220-1 v3.1.1 §5.18.6.4	Test Date(s):	5/17/2017
Declarations:	None		
Configuration:	1		
Test Setup:	The EUT is placed on Styrofoam platform at 1.5m in height. The input voltage is 230Vac/50Hz. USB and DC ports are unterminated and not accessible to user. The EUT is tested in preferred orientation declared by the manufacturer. The EUT is set into receive mode and paired with a companion device. A support signal generator is used to transmit unwanted signal and connected to a dipole antenna. Operating frequency: 433.92MHz Antenna gain: 0dBi Software output power: <1mW Test signal: D-M3 Frequency range of measurement = 433.92MHz +/-(5%*433.92)MHz RBW = 100 kHz, VBW = 100 kHz The manufacturer declares that the EUT is category 3 receiver.		

Environmental Conditions			
Ambient Temperature (ºC)	18	Relative Humidity (%):	42

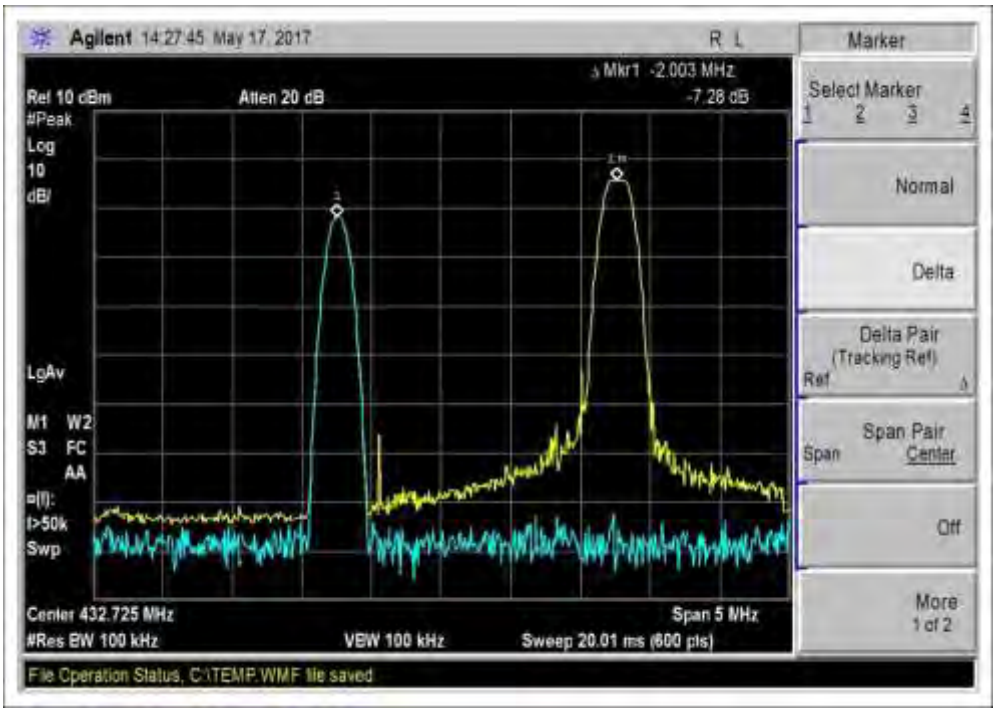
Test Equipment					
Asset#	**Description**	**Manufacturer**	**Model**	**Cal Date**	**Cal Due**
00309	Preamp	HP	8447D	3/14/2016	3/14/2018
01995	Biconilog Antenna	Chase	CBL6111C	5/10/2016	5/10/2018
P05275	Attenuator	Weinschel	1W	5/5/2016	5/5/2018
P05050	Cable	Pasternack	RG223/U	1/20/2017	1/20/2019
P05198	Cable	Belden	8268	12/7/2016	12/7/2018
02672	Spectrum Analyzer	Belden	E4446A	3/2/2017	3/2/2019
00786	Preamp	HP	83017A	5/9/2016	5/9/2018
03419	Signal Generator	Agilent	E4438C	7/8/2015	7/8/2017
P05497	Dipole Antenna	Schwarzbeck	UHAP	1/19/2017	1/19/2019

Test Data

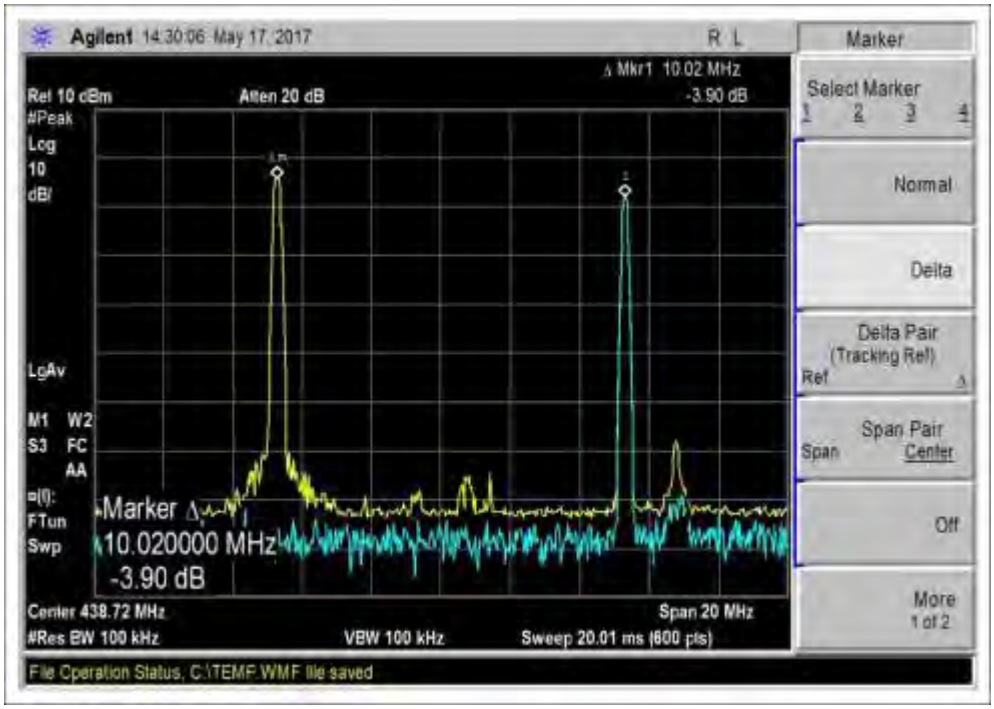
Operating Frequency (MHz)	Offset Frequency (MHz)	Signal Generator A Power (dBm)	Blocking Level (dBm)	Limit (dBm)	Result
433.92	+2	-19.19	-26.22	≥ -80	Pass, no degradation in performance
433.92	-2	-19.19	-26.47	≥ -80	Pass, no degradation in performance
433.92	+10	-19.19	-23.09	≥ -60	Pass, no degradation in performance
433.92	-10	-19.19	-28.07	≥ -60	Pass, no degradation in performance
433.92	+5%*433.92	-19.19	-23.89	≥ -60	Pass, no degradation in performance
433.92	-5%*433.92	-19.19	-30.54	≥ -60	Pass, no degradation in performance



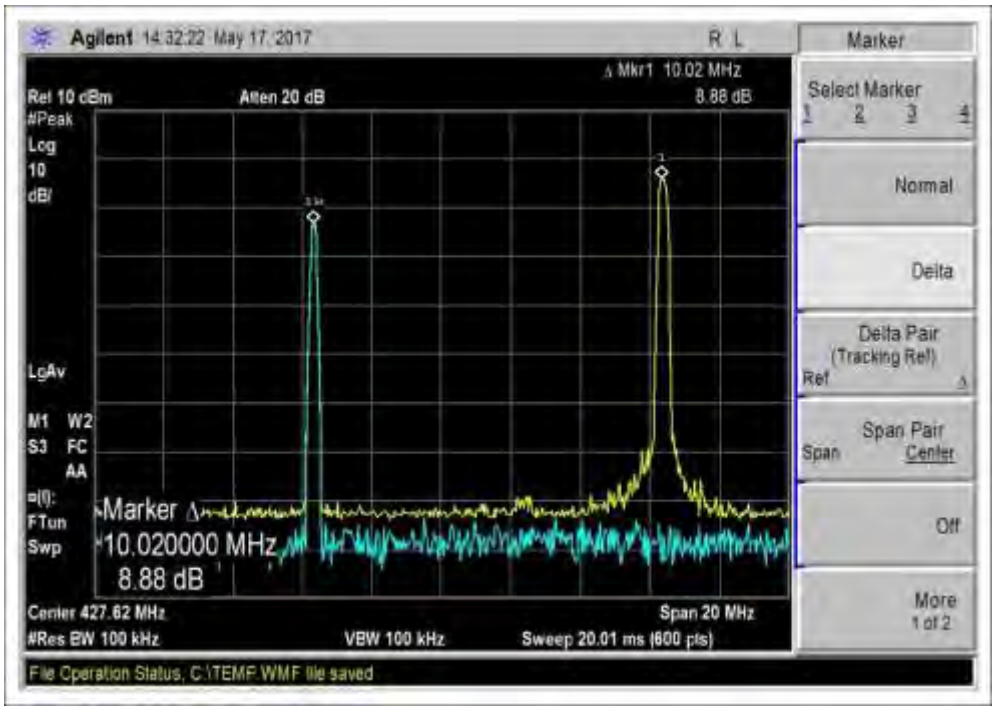
Blocking +2MHz



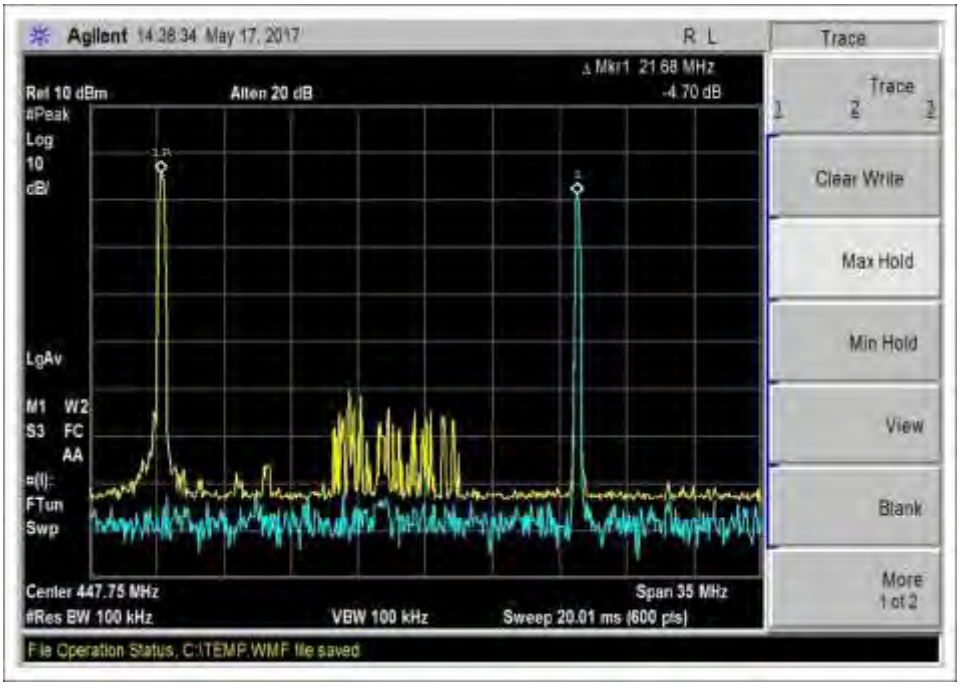
Blocking -2MHz



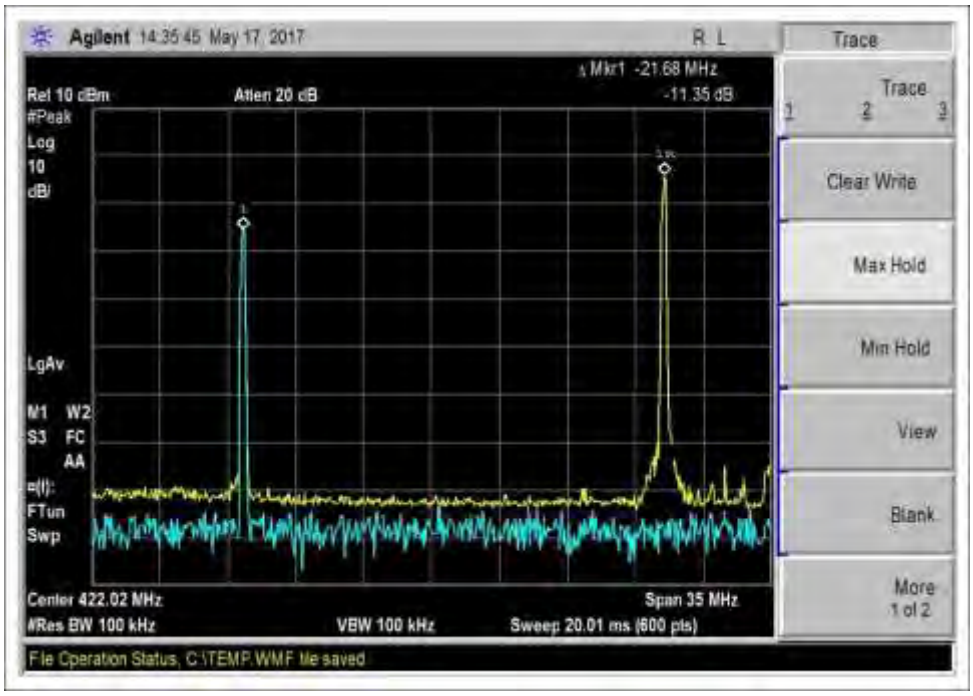
Blocking +10MHz



Blocking -10MHz



Blocking +5%*433.92MHz



Blocking -5%*433.92MHz

Test Setup Photo



SUPPLEMENTAL INFORMATION

Measurement Uncertainty

Parameter Uncertainty	Actual	Limit	Unit of Measure
Uncertainty Parameter	**Actual**	**Limit**	**Unit of Measure**
Occupied Channel Bandwidth	1	5	%
RF output power, conducted	0.67	1.5	dB
Power Spectral Density, conducted	0.67	3	dB
Unwanted Emissions, conducted	0.67	3	dB
All emissions, radiated	3.73	6	dB
Temperature	1	3	ºC
Humidity	3.4	5	%
DC and low frequency voltages	2	3	%
Time	1.1	5	%

Uncertainties reported are worst case for all CKC Laboratories' sites and represent expanded uncertainties expressed at approximately the 95% confidence level using a coverage factor of k=2.
Compliance is deemed to occur provided measurements are below the specified limits.